

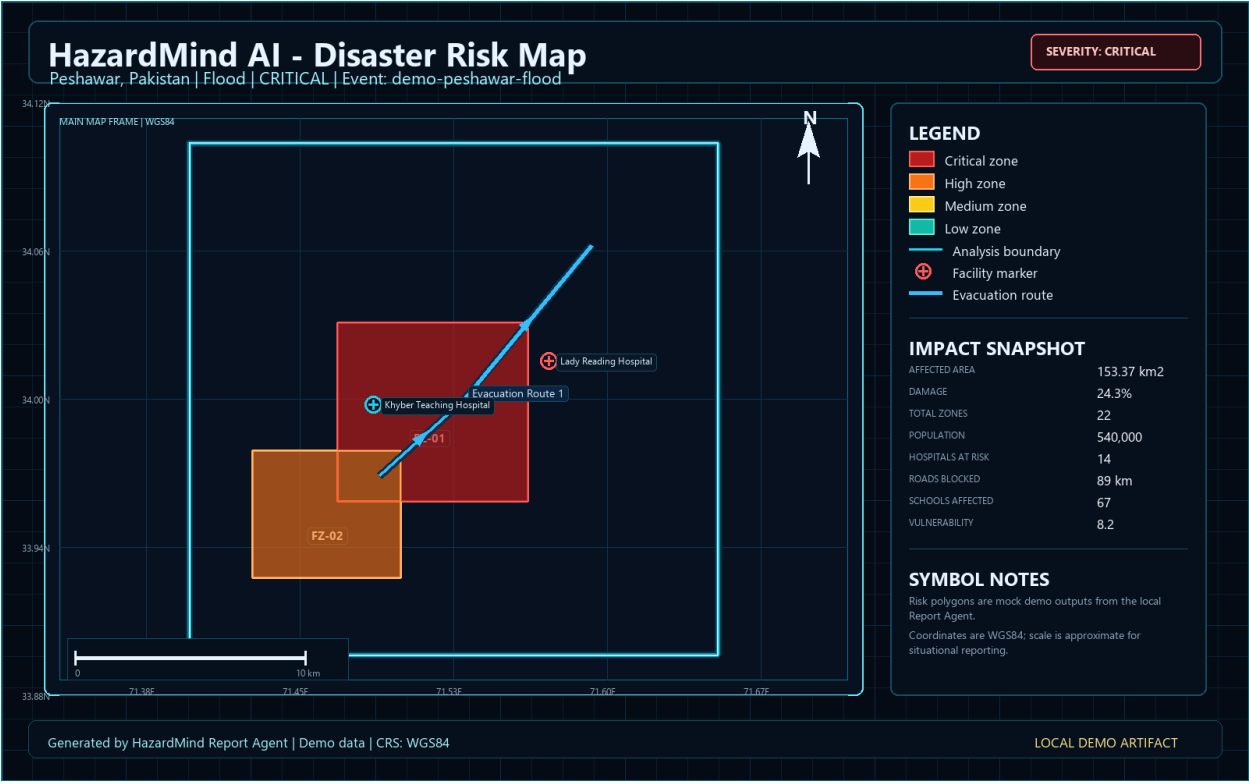
HazardMind AI Executive Disaster Report

Event ID	demo-peshawar-flood
Location	Peshawar, Pakistan
Hazard Type	Flood
Overall Severity	CRITICAL
Satellite Type	sentinel-1
Satellite Reason	cloud_cover_above_30_percent_sar_selected

Operational Statistics

Affected Area	153.37 km2
Damage Percent	24.3%
Total Zones	22
Population Affected	540,000
Hospitals at Risk	14
Roads Blocked	89 km
Schools Affected	67
Vulnerability Score	8.2

Generated Risk Map



Executive Summary

A CRITICAL flood in Peshawar, Pakistan has inundated approximately 153 km² across 22 hazard zones, with roughly 540,000 people affected and a high composite vulnerability score of 8.2. Detection was confirmed via Sentinel-1 SAR imagery (selected over optical sensors due to 42% cloud cover), yielding a 0.91 flood confidence; the most severe inundation is concentrated in zones FZ-01 and FZ-02 in the central-west sector. Critical infrastructure exposure is significant, with 14 hospitals at risk, 67 schools affected, and 89 km of roadway blocked, including HIGH-risk Lady Reading Hospital. Immediate priorities are life-safety evacuation and rescue from deep-water zones, hospital continuity-of-care planning, and opening additional evacuation corridors to relieve the single mapped route serving a large displaced population. Findings are model-derived and require urgent ground-truth validation, as SAR classification in dense urban areas carries inherent uncertainty.

Detailed Incident Analysis

Event demo-peshawar-flood describes a CRITICAL flood affecting Peshawar, Pakistan, detected and validated using Sentinel-1 SAR imagery (scene S1A_DEMO_PESHAWAR_20260613). SAR was selected over optical sensors because cloud cover reached 42 percent, exceeding the 30 percent threshold at which optical analysis becomes unreliable; SAR's all-weather, day-night radar capability provides dependable flood mapping under these conditions. Analysis indicates approximately 153.37 km² of inundated terrain across 22 delineated hazard zones, with an estimated 24.3 percent damage footprint within the assessed area. An estimated 540,000 people are affected. Critical infrastructure exposure is significant: 14 hospitals are at risk, 89 km of roadway are blocked, and 67 schools are affected. The composite vulnerability score is 8.2 on a 10-point scale, reflecting high population density, constrained mobility, and concentrated critical facilities in flood-exposed areas. The most severe inundation is concentrated in zone FZ-01 (deep_water, 12.4 km², critical severity) along the 71.47-71.56E / 33.96-34.03N corridor, with high-severity standing water in zone FZ-02 (8.7 km²). Lady Reading Hospital is flagged HIGH risk and Khyber Teaching Hospital MEDIUM risk, both requiring continuity-of-care contingency planning. One primary evacuation corridor (Evacuation Route 1) has been identified running northeast from 71.49E/33.97N toward higher ground near 71.59E/34.06N.

Technical Analysis

Flood detection relied on Sentinel-1 SAR using a VV/VH polarization ratio index with a mean value of 0.24, consistent with widespread open-water and saturated surfaces (low backscatter returns characteristic of smooth water bodies). The hazard model assigns a CRITICAL flood risk with 0.91 confidence, materially above the secondary hazards of earthquake (MEDIUM, 0.67 confidence) and landslide (LOW, 0.54 confidence). The high flood confidence combined with the deep_water classification in FZ-01 suggests sustained, not transient, inundation. The 24.3 percent damage estimate across 153.37 km² implies roughly 37 km² of directly impacted built and natural surface. The vulnerability score of 8.2 amplifies operational concern: with 540,000 people exposed and 89 km of arterial routes blocked, access degradation will compound medical and logistics constraints. The clustering of critical zones FZ-01 and FZ-02 in the central-west sector overlaps the service catchments of both named hospitals, indicating potential simultaneous demand surge and access loss. SAR-based estimates carry inherent classification ambiguity in urban environments due to double-bounce and layover effects, which should temper precise per-building damage interpretation. The single mapped evacuation route represents a capacity bottleneck given the affected population scale.

Recommendations

- Immediately activate flood emergency operations and surge command for the Peshawar central-west sector covering zones FZ-01 and FZ-02.
- Initiate continuity-of-operations and patient-evacuation contingency for Lady Reading Hospital (HIGH risk) and pre-position transfer capacity for Khyber Teaching Hospital (MEDIUM risk).
- Deploy rapid ground assessment teams to validate SAR-derived deep_water zones, prioritizing FZ-01 before committing rescue assets.
- Identify and open at least two additional evacuation corridors to supplement Evacuation Route 1 and prevent single-route saturation.
- Clear or bypass the 89 km of blocked roadway in priority order linking population centers to the northeast high-ground corridor.
- Establish shelter, potable water, and sanitation capacity scaled to the 540,000 affected population to mitigate secondary public-health risks.
- Issue targeted public warnings and movement guidance for residents within and adjacent to mapped critical and high-severity zones.

Response Priorities

- Life safety: evacuation and rescue from FZ-01 deep_water and FZ-02 high-severity zones.
- Healthcare continuity: protect and decompress the 14 at-risk hospitals, starting with HIGH-risk facilities.
- Access restoration: reopen or establish viable routes across the 89 km of blocked roads.
- Population care: shelter, water, sanitation, and medical support for displaced residents.
- Situational awareness: validate satellite findings with ground truth and refresh imagery.

Assumptions

- Population, hospital, road, and school impact figures reflect current exposure within mapped hazard zones.
- SAR VV/VH classification accurately distinguishes open water and saturated surfaces from dry terrain.
- Evacuation Route 1 remains passable and leads to safe high ground at the northeast terminus.
- Secondary earthquake and landslide risks remain stable during the flood response window.

- Critical facility coordinates and risk ratings are current and accurate.

Limitations

- Cloud cover of 42 percent precluded optical verification, leaving SAR as the sole primary sensor.
- SAR backscatter in dense urban areas is subject to layover and double-bounce effects that can misclassify flood extent.
- Only 2 of 22 zones are detailed in the provided feature set, limiting full spatial granularity.
- Damage percent and affected-area values are model estimates pending ground-truth confirmation.
- Only one evacuation route is mapped, which may understate or overstate true egress capacity.
- Confidence scores reflect model certainty, not field-verified conditions.

Model Source Note

Featherless detailed generation failed; AI/ML fallback was used. Executive summary generated using AI/ML API.

Detailed Report	aiml_fallback
Executive Summary	aiml
Fallback Used	True
Featherless Model	moonshotai/Kimi-K2.6

Agent Log / Trace

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:00Z	Report Agent received event context
hazardmind-report	complete	2026-06-13T18:03:20Z	Featherless Kimi K2.6 unavailable, AI/ML fallback used
hazardmind-report	complete	2026-06-13T18:03:40Z	AI/ML generated executive summary
hazardmind-report	complete	2026-06-13T18:04:00Z	Static cartography map generated locally